MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548007

(GENERATOR, COMPUTER TIMING)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the computer timing generator.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies General Specification for

DRAWINGS

U. S. Army: Frankford Arsenal

F10548007	Computer Timing Generator

PACKAGING DATA SHEET

10548007	Packaging of Computer Timing Generator

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548007 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram for the circuit card assembly.

3.5 Performance. - When supplied with the input power and signals given in Table I the computer timing generator shall yield the output signals as specified in Table II and III. Signals specified as logic levels or composed of logic levels shall have a logic 1 level +3.5 ±1.0 volts (V) and for a logic 0 level 0 +0.5 V.

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TABLE I. - INPUT SIGNAL CHARACTERISTICS

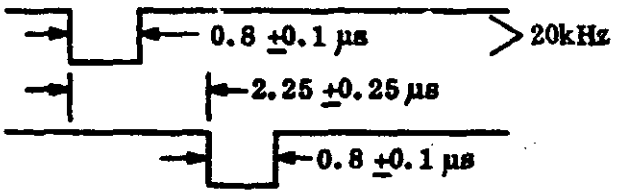
SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
Input power	5.0 ±0.3 Vdc, regulated; ac ripple, less than 50 mV peak-to-peak; capacity 300 mA.	P1-30, P1-31, & P1-32 (+5V) P1-24, P1-25, & P1-26 (+5V RET)
Input signals		
$\overline{T1}$		P1-22
T3		P1-15
TARGET DATA	Logic 1 when active.	P1-9
R200/T2	Not normally used except for testing of assembly by applying TG SET as an input.	P1-14
MANUAL TARGET REJECT	Logic 0 when active.	P1-20
R MAX	Logic 0 when active.	P1-34

TABLE II. - OUTPUT SIGNAL CHARACTERISTICS

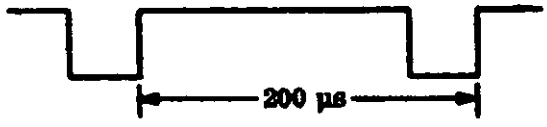
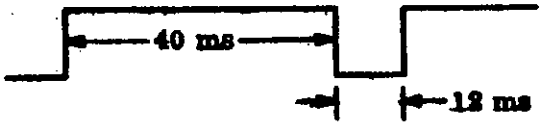
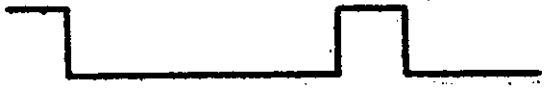
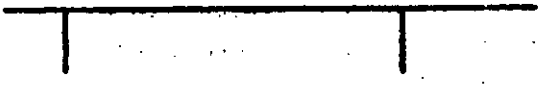
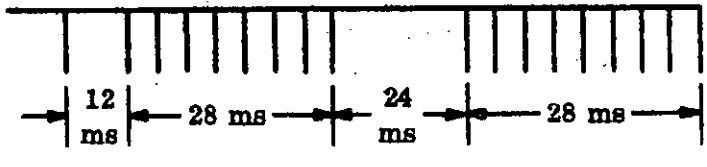
SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
Output signals		
T1	Complement of $\overline{T1}$ 	P1-10
	Complement of $\overline{T3}$ >20 kHz	
T3		P1-16
T3/4KC	Logic 1 pulse being coincident with T3 and having the same characteristics but occurring once for every five T3 pulses.	P1-19
T1/250	Logic 0 pulse being coincident with T1 and having the same characteristics but occurring every 4.0 ±0.25 ms.	P1-18

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TABLE III. - OUTPUT SIGNAL WAVEFORMS

SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS

For F4/5 through F1/5 each division is equal to 50 µs.

F4/5		P1-45
$\overline{F4/5}$		P1-47
F2/5		P1-4
$\overline{F2/5}$		P1-2
F1/5		P1-13
$\overline{F1/5}$		P1-11
200M		P1-50
$\overline{200M}$		P1-55
$\overline{TG\ SET}$		P1-7
$\overline{TG\ RESET}$		P1-5

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification of the circuit card assembly.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the test equipment listed in Table IV shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE IV. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2. Pulse generator	Frequency: 20 kHz minimum. Duty factor: 10% minimum. Amplitude: Minimum of 4 V into a 50-ohm load. Pulse rise and fall times less than 0.1 ms.
3. Digital voltmeter	With accuracy and resolution of ±0.2% or better from 0.0090 to 99.0 Vdc. Input impedance 10 megohms or greater on all voltage scales.
4. Test fixture	See Figure 1. See Table V for test fixture parts list.
5. Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per division or greater, input resistance greater than 8 megohms, input capacitance less than 25 PF.

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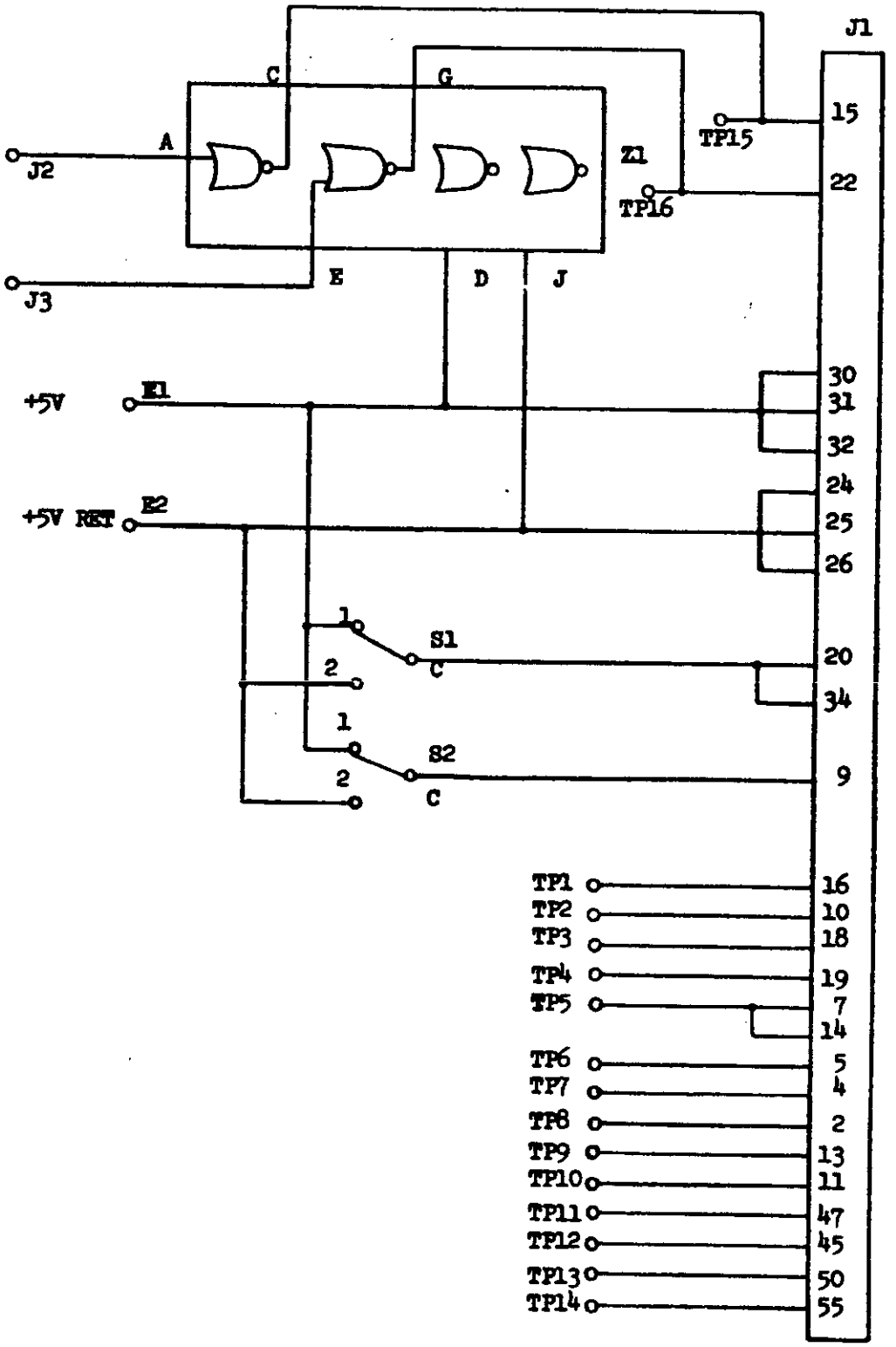


FIGURE 1. TEST FIXTURE SETUP.

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TABLE V. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptable, UG-625B/U or equivalent.
J3	Same as J2.
Z1	Integrated circuit, SG-140-03, Sylvania or equivalent.
S1 & S2	Switch, SPDT.
TP1 thru TP16	Tip jack, H. H. Smith 1501 or equivalent.

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4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected computer timing generators shall be tested as follows: Adjust power supply, item 1 (see Figure 2), and pulse generators, item 2, to delivery signals as shown in Figure 3, "Fixture Input Signals". Connect test equipment, items 1 through 4, as shown in Figure 2. Insert computer timing generator into J1 in test fixture, item 4.

4.6.3.1 $\overline{T1}$ and $\overline{T3}$ input. - Using channels A and B of oscilloscope observe input signals labeled as $\overline{T1}$ and $\overline{T3}$ on Table I on TP16 and TP15 respectively. Synchronize oscilloscope on TP16.

4.6.3.2 T1 and T3 output. - Using channel A and B of oscilloscope observe output signals labeled at T1 and T3 on Table II on TP2 and TP5 respectively. Synchronize oscilloscope on TP16.

4.6.3.3 T3/4KC output. - Connect channel A of oscilloscope to TP4 and channel B to TP5. Synchronize oscilloscope on TP4. Observe pulse on channel A with characteristics same as output signal labeled T3 on Table II. There shall also be five pulses on channel B for each pulse on channel A.

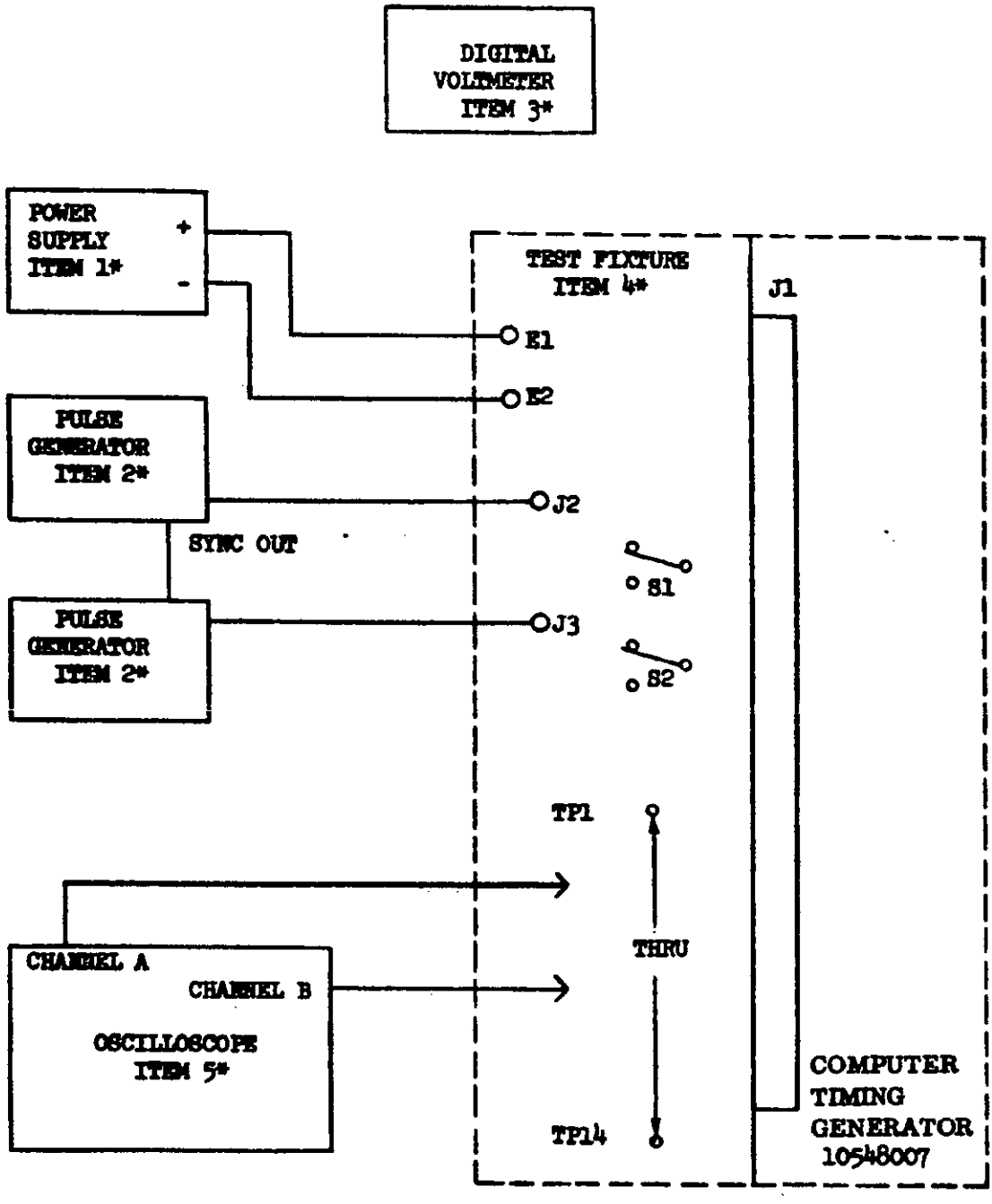
4.6.3.4 $\overline{T1/250}$ output. - Connect channel A of oscilloscope to TP3 and channel B to TP2. Synchronize oscilloscope to TP3. Observe pulse on channel A with characteristics same as input signal labeled $\overline{T1}$ on Table II and be coincident with one pulse on channel B. Pulses on channel A shall be spaced every 4 microseconds (μs).

4.6.3.5 F4/5 and $\overline{F4/5}$ output. Connect channel A of oscilloscope to TP12 and channel B to TP11. Synchronize oscilloscope on TP4. Observe output signals labeled as F4/5 and $\overline{F4/5}$ on Table III.

4.6.3.6 F2/5 and $\overline{F2/5}$ output. - Connect channel A of oscilloscope to TP7 and channel B to TP6. Synchronize oscilloscope on TP4. Observe output signals labeled as F2/5 and $\overline{F2/5}$ on Table III.

4.6.3.7 F1/5 and $\overline{F1/5}$ output. - Connect channel A of oscilloscope to TP9 and channel B to TP10. Synchronize oscilloscope on TP4. Observe output signals labeled as F1/5 and $\overline{F1/5}$ on Table III.

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* REFER TO TABLE IV

FIGURE 2. COMPUTER TIMING GENERATOR SETUP.

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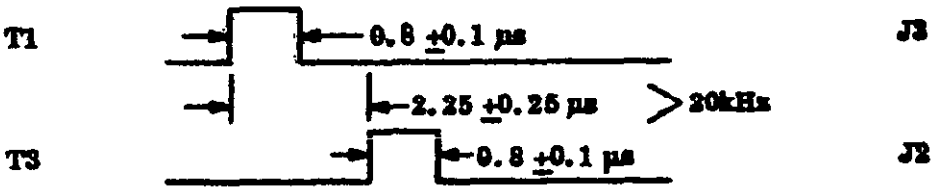


FIGURE 3. FIXTURE INPUT SIGNALS.

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4.6.3.8 200 meter counter test. - Connect channel A of oscilloscope to TP13 and channel B to TP14. Synchronize oscilloscope on TP13. With S1 and S2 in position 1 observe output signals labeled as 200M and $\overline{200M}$ on TP13 and TP14 respectively as per Table I. When S1 is placed to position 2 a logic 1 shall be observed on TP13 and a logic 0 on TP14. Return S1 to position 3.

4.6.3.9 Signal conditioner test. - Connect channel A of oscilloscope to TP1 and Channel B to TP6. Synchronize oscilloscope to TP13. Observe output signals labeled as $\overline{\text{TG SET}}$ and $\overline{\text{TG RESET}}$ on TP1 and TP13 respectively as shown on Table III.

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TABLE VI. - CROSS REFERENCE FROM SIGNAL TO TEST POINTS

SIGNAL	BOARD TERMINALS	TEST POINTS
$\overline{\text{T1}}$	P1-22	TP16
$\overline{\text{T3}}$	P1-15	TP15
T1	P1-10	TP2
T3	PL-16	TP5
T3/4KC	P1-19	TP4
$\overline{\text{T1/250}}$	P1-18	TP3
F4/5	T1-45	TP12
$\overline{\text{F4/5}}$	P1-47	TP11
F2/5	P1-4	TP7
$\overline{\text{F2/5}}$	P1-2	TP8
F1/5	P1-13	TP9
$\overline{\text{F1/5}}$	P1-11	TP10
200M	P1-50	TP13
$\overline{\text{200M}}$	P1-55	TP14
$\overline{\text{TG SET}}$	P1-7	TP1
$\overline{\text{TG RESET}}$	P1-5	TP6

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548007. The level of protection shall be as specified in procurement document.

6. NOTES

6.1 Intended use. - The computer timing generator is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The computer timing generator uses binary counters in cascade to produce timing pulses of 4 kHz, 250 Hz, and 25 Hz. Also provided by this assembly is a signal conditioner and a five-line decoder output.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A239

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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